Preliminary Design for 75 m Offshore Patrol Vessel Option in Guatemala

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ABSTRACT

This study examines a 75 m monohull Offshore Patrol Vessel as a preliminary option for Guatemalan maritime missions. The analysis uses the selected principal dimensions, 1,500 ton design displacement, 285.51 ton fuel capacity, 1264.68 BHP at 14 kn, and 11300.55 BHP at 30 kn. The dimensional data are checked for geometric displacement consistency. The ITTC-1957 correlation line is applied separately to calculate frictional resistance at the two operating conditions. A power-law BHP relation is used to display values at 2 kn increments from the stated 14 kn and 30 kn conditions. Fuel consumption, endurance, and range are calculated from the stated specific fuel-consumption values, a declared 501 kW auxiliary-load input, a 10% consumption margin, and a separate 10% fuel reserve. At 14 kn, the calculated fuel rate is 809.70 lb/hr, endurance is 29.15 days, and range is 9795 nm. At 30 kn, the corresponding values are 5004.69 lb/hr, 4.72 days, and 3396 nm. The resulting analysis provides a transparent basis for comparing sustained-patrol and high-speed-response conditions.

Keywords: Offshore Patrol Vessel; brake power; fuel consumption; endurance range; ITTC-1957.

1. Introduction

This paper examines a 75 m monohull Offshore Patrol Vessel option for Guatemala, defined by a 14 kn cruise condition and a 30 kn maximum-speed condition. It evaluates the power, fuel-consumption, endurance, and range implications of those two stated operating conditions. The calculation sequence is deliberately restricted to parameters whose inputs, equations, units, and numerical outputs are displayed in the paper. The treatment of resistance, propulsion, and power follows established naval-architecture references [1–5]. The operational rationale is the need for a patrol platform able to reach and maintain presence across a maritime area extending to 200 nautical miles (nm) from the baselines, the maximum breadth of the exclusive economic zone under Article 57 of the United Nations Convention on the Law of the Sea [11]. For Guatemala, the intended roles of the proposed option are maritime sovereignty presence, surveillance, fisheries protection, support to interdiction of illicit maritime activity, and search-and-rescue support. The SAR Convention identifies coordinated rescue services and the organization of available resources as essential elements of maritime search and rescue [12]. The proposed operational concept is intended for an embarked complement of 30–50 personnel. This is an accommodation requirement for the OPV proposal; it is not used to infer generator size, freshwater storage, structural weights, or compartment arrangement. Those items require a general arrangement and machinery definition, whereas the present study evaluates the

declared 501 kW auxiliary-load input, propulsion power, fuel use, endurance, and range. This separation is consistent with naval-architecture practice, in which hull-form, propulsion, and ship-integration work are developed as linked but distinct design disciplines [6].

2. Materials and Methods

2.1 Design Inputs and Geometric Displacement Consistency

The following calculation checks the internal geometric (shown in Table 1) consistency of the selected dimensions and block coefficient. It is not a stability calculation. Seawater density is taken as 1.025 ton/m³.

$$\nabla = LWL \cdot B \cdot T \cdot CB = 75.0 \cdot 11.2 \cdot 3.4 \cdot 0.5123 = 1{,}463.1288\ m^3$$

$$\Delta calc = \rho \nabla = 1.025 \cdot 1{,}463.1288 = 1{,}499.707\ t$$

The geometric calculation gives 1,499.707 ton; the selected design displacement is therefore reported as 1,500 ton after rounding to the nearest ton. This check only confirms that the stated length, beam, draft, and block coefficient are mutually consistent.

Table 1. Selected principal dimensions and calculation

Parameter	Symbol	Value	Unit
Length at waterline	LWL	75.00	m
Module beam	B	11.20	m
Module depth	D	6.20	m
Design draft	T	3.40	m
Block coefficient	CB	0.5123	-
Design displacement	Δ	1,500	ton
Fuel capacity	M_fuel	285.51	ton
Wetted surface area	S	1,117.5	m^2
Seawater density	ρ	1,025	kg/m^3
Kinematic viscosity	ν	1.37×10^{-6}	m^2/s

2.2 Operating Conditions and BHP Interpolation

The BHP values at 14 kn and 30 kn are the stated design conditions. Figure 1 and Table 4 report a calculated BHP value at every 2 kn. Values from 14 kn to 30 kn are obtained by interpolation between the two stated conditions; values below 14 kn are displayed as the mathematical extension of the same equation.

All numerical calculations and the BHP-speed plot were prepared in Microsoft Excel from the declared inputs and equations presented in this study (shown in Table 2).

$$n = \ln(BHP30 / BHP14) / \ln(30 / 14) = 2.8735$$

$$BHP(Vkn) = BHP14 \cdot (Vkn / 14)^n$$

Table 2. Declared operating and fuel-calculation inputs.

Item	14 kn	30 kn
Brake power (BHP)	1,264.68	11,300.55
Auxiliary electrical load (kW)	501	501
Main-engine SFOC (lb/BHP·hr)	0.380	0.380
Auxiliary SFOC (lb/kW·hr)	0.510	0.510
Fuel reserve	10%	10%
Consumption margin	10%	10%

2.3 Frictional Resistance at the Stated Conditions

The ITTC-1957 correlation line is used only to calculate the bare-hull frictional resistance at 14 kn and 30 kn [1,2]. It is not identified as total resistance (shown in Table 3).

$$Re = VLWL / \nu$$

$$C_F = 0.075 / (\log 10\ Re - 2)^2$$

$$R_F = 0.5 \rho V^2 SCF$$

$$Fr = V / \sqrt{(gLWL)}$$

The Froude number is included as a dimensionless speed parameter. It is reported to identify the nondimensional operating condition; it is not used to convert frictional resistance into total resistance.

Table 3. ITTC-1957 frictional-resistance calculation and Froude number

Quantity	14 kn	30 kn
Speed (m/s)	7.202	15.433
Reynolds number	3.943×10^8	8.449×10^8
Froude number, Fr	0.2656	0.5691
Friction coefficient, C_F	0.001724	0.001563
Frictional resistance, RF (kN)	51.22	213.23

2.4 Fuel, Endurance, and Range Equations

Fuel consumption is calculated at system level from the declared brake power and auxiliary electrical load. The 10% consumption margin and the 10% fuel reserve are separate quantities: the margin increases hourly consumption once, whereas the reserve reduces the fuel available for the mission.

$$\dot{m}fuel = (BHP \cdot SFOCME + Paux \cdot SFOCAE)(1 + fm)$$

$$Musable = Mtank(1 - reserve)$$

$$tend = Musable \cdot 2{,}204.62262 / \dot{m}fuel; \quad Range\ (NM) = Vkn \cdot tend\ (h)$$

3. Results

3.1 BHP-Speed Trend

Figure 1 and Table 4 display the BHP relation defined by the displayed power-law calculation. Each marker corresponds to a 2 kn increment and carries its calculated BHP value. The highlighted points at 14 kn and 30 kn are the stated design conditions. The solid 14–30 kn segment is an interpolation; the dashed 2–14 kn segment is the mathematical extension of the same relation.

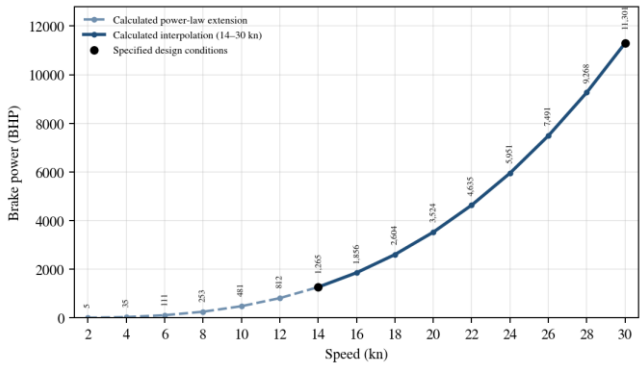


Figure 1. Speed vs. BHP diagram at 14 kn and 30 kn

The values in Table 4 reproduce the labels in Figure 1. This arrangement makes the BHP-speed relation readable both in graphical form and as numerical values for direct checking or spreadsheet use.

Table 4. Calculated Speed vs. BHP table

Speed (kn)	BHP (kW)	Speed (kn)	BHP (kW)
2	4.72	18	2,603.82
4	34.56	20	3,524.49
6	110.81	22	4,634.89
8	253.28	24	5,951.49
10	480.93	26	7,490.58
12	812.10	28	9,268.28
14	**1,264.68**	**30**	**11,300.55**
16	1,856.19		

Table 5 uses the same specific fuel-consumption values and the same 501 kW auxiliary load at every speed. The total fuel column applies the 10% consumption margin once to the sum of main-engine and auxiliary fuel use.

Table 5. Calculated fuel-consumption values

Speed (kn)	BHP (kW)	Main fuel (lb/hr)	Aux. fuel (lb/hr)	Total fuel with 10% margin (lb/hr)
2	4.72	1.79	255.51	283.03
4	34.56	13.13	255.51	295.51
6	110.81	42.11	255.51	327.38
8	253.28	96.25	255.51	386.93
10	480.93	182.75	255.51	482.09
12	812.10	308.60	255.51	620.52
14	1,264.68	480.58	255.51	809.70
16	1,856.19	705.35	255.51	1,056.95
18	2,603.82	989.45	255.51	1,369.46
20	3,524.49	1,339.31	255.51	1,754.30
22	4,634.89	1,761.26	255.51	2,218.44
24	5,951.49	2,261.57	255.51	2,768.78
26	7,490.58	2,846.42	255.51	3,412.13
28	9,268.28	3,521.95	255.51	4,155.20
30	11,300.55	4,294.21	255.51	5,004.69

3.2 Fuel Consumption, Endurance, and Range

At 14 kn, the fuel calculation is (1,264.68 × 0.380 + 501 × 0.510) × 1.10 = 809.70 lb/hr. With usable fuel of 0.90 × 285.51 = 256.959 ton, the calculated endurance and range are 699.6 hr, 29.15 days, and 9795 nm. At 30 kn, the same equation gives (11,300.55 × 0.380 + 501 × 0.510) × 1.10

= 5004.69 lb/hr. The same usable fuel mass gives 113.2 hr, 4.72 days, and 3396 nm. (shown in Table6)

Table 6. Fuel, endurance, and range at the two stated operating conditions

Quantity	14 kn	30 kn
Main-engine fuel rate (lb/hr)	480.58	4,294.21
Auxiliary fuel rate (lb/hr)	255.51	255.51
Fuel rate incl. 10% margin (lb/hr)	809.70	5,004.69
Usable fuel after 10% reserve (ton)	256.959	256.959
Endurance (hr)	699.6	113.2
Endurance (day)	29.15	4.72
Range (nm)	9,795	3,396

3.3 Calculation Traceability and Unit Control

Each reported result is obtained from a declared input or from a preceding calculation in this paper. Speed is converted from knots to metres per second before the Reynolds-number, Froude-number, and frictional-resistance calculations. Fuel is retained in tonnes for tank capacity and is converted to pounds only when it is divided by a fuel rate expressed in lb/h. This prevents a unit mismatch between storage mass, fuel rate, endurance, and range. An arithmetic audit was applied to the displayed calculation chain. The dimensional product gives 1,463.1288 m^3, and multiplication by 1.025 ton/m^3 gives 1,499.707 ton before rounding. At 14 kn, the speed conversion gives 7.202 m/s; with LWL = 75.00 m and $\nu = 1.37 \times 10^{-6}$ m^2/s, the Reynolds number is 3.94×10^8. The ITTC-1957 correlation then gives $C_F = 0.001724$ and $R_F = 51.22$ kN. The corresponding 30 kn values are 15.433 m/s, 8.45×10^8, 0.001563, and 213.23 kN. The Froude-number values are 0.2656 at 14 kn and 0.5691 at 30 kn. They are calculated directly from speed, gravitational acceleration, and waterline length. They are included to describe the nondimensional speed range, whereas the reported resistance remains the ITTC frictional component calculated from the wetted surface and friction coefficient. The calculation uses a 10% fuel reserve as a deduction from tank capacity: 0.90 × 285.51 ton = 256.959 ton. The consumption margin is applied to the hourly fuel rate, not to the fuel capacity. Thus the reserve and the margin are not combined or counted twice. The same convention is used at both speeds, which makes their endurance and range directly comparable.

4. Engineering Interpretation

4.1 Displacement Check

For a preliminary dimensional check, displacement volume is represented by the product of waterline length, beam, draft, and block coefficient. The calculated volume is 1,463.13 m^3. At the adopted seawater density, the geometric displacement is 1,499.7 ton, which rounds to the selected 1,500 ton design displacement. This calculation confirms compatibility among the selected numerical inputs and gives the mass basis used for the stated vessel concept. It does not require a general arrangement or a detailed weight estimate because it is strictly a volumetric consistency calculation.

4.2 Speed-Power Relation

The BHP curve has an exponent of 2.8735, derived directly from the ratio of the two specified BHP values and the ratio of their speeds. It therefore passes exactly through 1,264.68 BHP at 14 kn and 11,300.55 BHP at 30 kn. The calculated intermediate points are suitable for displaying the mathematical trend, for checking that a plotted curve is continuous, and for making spreadsheet calculations consistent with the two stated conditions. The curve should be interpreted as an interpolation within the stated 14–30 kn operating envelope. It is not a substitute for a resistance test, model basin campaign, CFD study, or sea-trial record. Its technical purpose in this paper is limited and explicit: it connects the two declared power conditions through a single reproducible equation, avoiding unrelated or manually entered BHP values at intermediate speeds.

4.3 Friction, Power, and Fuel Results

The frictional-resistance calculation and the BHP calculation serve different roles. The ITTC expressions give the frictional component from speed, water properties, wetted surface, and the correlation line. The power-law

expression gives the stated BHP trend. Keeping the two result sets separate avoids treating frictional resistance as total ship resistance. In particular, the 51.22 kN result at 14 kn and the 213.23 kN result at 30 kn are named and tabulated as frictional-resistance values only. Fuel use rises sharply with the BHP requirement. Although the 501 kW auxiliary load is held constant in the stated model, main-engine fuel consumption increases with BHP while the fuel inventory remains fixed. The two operating conditions therefore provide different, complementary operating roles rather than interchangeable endurance conditions.

4.4 Operational Relevance of the Two Conditions

Within the stated model, 14 kn is the sustained-patrol condition: its lower BHP requirement preserves fuel and produces the greater calculated endurance and range. This condition is relevant to routine maritime presence, surveillance, search-and-rescue readiness, fisheries protection, and other patrol activities in which time on station is more important than maximum speed. At 14 kn, a 200 nm transit takes approximately 14.3 hr; a 200 nm outward transit and return represent 400 nm, or about 4.1% of the calculated 9,795 nm range. These figures do not establish an operational patrol plan, but they show that the stated cruise condition provides a numerical basis for transit to the outer part of a 200 nm maritime area and continued presence under the model assumptions. The 30 kn condition is the high-speed-response condition. At that speed, a 200 nm transit takes approximately 6.7 hr instead of 14.3 hr. Its value is therefore response time rather than long endurance. The numerical comparison illustrates a practical operational trade-off: the same vessel concept can support sustained patrol at 14 kn and rapid response at 30 kn, with the associated fuel penalty made explicit by the calculation. The monohull designation is used here as a single, consistent baseline for the stated length, beam, draft, block coefficient, and wetted surface. This paper does not compare monohull performance with catamaran or trimaran alternatives; such a comparison would require additional hull-form data and a separate resistance analysis.

5. Verification and Engineering Scope

The numerical calculations were performed in Microsoft Excel using the declared inputs and equations reported in Sections 2 and 3. Unrounded values were retained during calculation, and the values in the tables were rounded only at the reporting stage. This procedure maintains consistency between the reported displacement, BHP, fuel rate, endurance, and range values. Unit control is explicit throughout the calculation: speed is converted from knots to metres per second for the Reynolds-number, Froude-number, and frictional-resistance calculations; BHP is paired with lb/BHP·h; auxiliary electrical load is paired with lb/kW·h; and usable fuel mass is converted from tonnes to pounds before it is divided by a fuel rate in lb/h.

The analysis is deliberately restricted to geometric displacement consistency, ITTC-1957 frictional resistance, the BHP-speed relation defined by the two stated operating conditions, and the resulting fuel-consumption, endurance, and range calculations. The frictional term is not presented as total resistance, and the BHP curve is not presented as a model-test or sea-trial prediction.

6. Conclusions

The selected dimensions and block coefficient give a geometric displacement of 1,499.7 ton, which rounds to the selected 1,500 ton design displacement. The corresponding ITTC-1957 frictional-resistance results are 51.22 kN at 14 kn and 213.23 kN at 30 kn. The stated operating conditions require 1,264.68 BHP at 14 kn and 11,300.55 BHP at 30 kn. The BHP-speed plot visualizes these conditions and provides calculated intermediate values through one explicit interpolation equation. Using 285.51 t fuel, a 501 kW auxiliary load, the specified fuel-consumption values, a 10% consumption margin, and a separate 10% fuel reserve, the calculated results are 29.15 days and 9795 nm at 14 kn, and 4.72 days and 3396 nm at 30 kn. For the stated Guatemala-focused mission rationale, the two speeds serve complementary purposes: 14 kn provides the calculated sustained-patrol condition for presence across a 200 nm maritime area, whereas 30 kn provides the calculated high-speed-response condition. The monohull is treated as a single baseline configuration,

not as a comparison with alternative hull forms. The study therefore provides a preliminary, internally consistent assessment of the proposed OPV option using one coherent chain of equations, units, data, and results.

Glossary

Symbol	Meaning
BHP	Brake power
C_B	Block coefficient
C_F	ITTC-1957 friction coefficient
Fr	Froude number
LWL	Length at waterline
Paux	Auxiliary electrical load
Re	Reynolds number
R_F	Frictional resistance
SFOC	Specific fuel-consumption rate
V	Speed
$\dot{m}_{fuel}$	Fuel-consumption rate
Δ	Displacement mass
∇	Displacement volume

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以瓜地馬拉 75 公尺船長近岸巡邏艦初步設計方案研究

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本研究以一艘 75 公尺單體近海巡邏艦為瓜地馬拉海上任務的初步方案進行分析。分析採用選定的主要尺寸參數：設計排水量 1500 噸，燃油容量 285.51 噸，14 節航速下功率 1264.68 馬力，30 節航速下功率 11300.55 馬力。對尺寸資料進行了幾何排水量一致性檢驗。分別應用 ITTC-1957 相關線計算兩種工況下的摩擦阻力。採用冪律功率關係式，以 2 節為增量，從上述 14 節及 30 節航速開始計算功率值。燃油消耗量、續航時間和航程根據公佈的比油耗值、501 千瓦輔助負載輸入、10%的油耗裕度和 10%的燃油儲備計算。在 14 節航速下，計算得到的燃油消耗率為 809.70 磅/小時，續航時間為 29.15 天，航程為 9,795 海浬。在 30 節航速下，對應的數值分別為 5004.69 磅/小時、4.72 天和 3396 海浬。此分析結果為比較持續巡邏和高速回應兩種情況下的續航時間和航程提供了清晰的依據。

關鍵字: 近岸巡邏艦，制動馬力，燃油消耗量，續航距離，1957 年國際拖行水槽會議